

Recovery on Signal or on Schedule?

A Small Pre-Registered Test of Health-Preserving Pedagogy in Continual Learning

Anton Dzyatkovskii 
Palo Alto AI Research Lab, Palo Alto, CA
dzyatkovskiy.a2@gmail.com

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Abstract

Health-preserving pedagogy holds that teaching has a cost to the learner and that a recovery phase should start when the learner’s diagnosed state calls for one, not when the timetable does. We test the machine analogue at a fixed gradient-step budget in class- and domain-incremental continual learning (5 tasks from the 8×8 digits set, a small MLP).

A threshold arm replays a small memory only when a held-out probe of past tasks shows a loss rise above τ . We compare it with replay of exactly the same count placed five other ways: uniformly over the whole run, at random, at the start of each task, at random inside each task, and evenly inside each task with the threshold arm’s own per-task counts. A sixth comparison is a sensor-free rule that makes replay proportional to the number of past tasks.

On a pre-registered fresh holdout (seeds 20–29) at equal cost, the threshold arm beats globally uniform replay. On class-incremental Split-Digits the gain is 3.4–7.7 points (8–10 of 10 seeds at τ 0.5–1.0); on Permuted-Digits it is 0.6–4.1 points and reaches 8–10 of 10 seeds only at $\tau \geq 0.75$. The sensor-free proportional rule is never beaten by the threshold arm at any of the ten points (the threshold arm is higher in 1–7 of 10 seeds) and exceeds it by 6.2 points at the smallest budget on Split. An oracle control that keeps the sensor’s per-task counts but spaces them evenly matches the threshold arm at medium budgets and exceeds it at the two smallest.

What transfers from the pedagogy is the dosage rule: more recovery as more material accumulates, spread evenly. On this benchmark the state alarm adds nothing measurable beyond what a fixed proportional rule already gives, and at small budgets its within-task timing costs accuracy. Code, seeds, raw tables, the pre-registration with two dated addenda and two superseded versions are public at <https://github.com/tonydz/machine-pedagogy>.

1 Motivation from pedagogy

Health-preserving pedagogy treats learning as an activity with a price [1, 2, 3]. The price includes fatigue, overload and loss of the capacity to keep learning, and it has to be paid back by recovery. The tradition’s working rules are about state: the teacher diagnoses the class (attention drop, error rise) and switches to a recovery or consolidation phase when the signal crosses a threshold, rather than at a fixed point of the timetable. In a study of more than 2000 schoolchildren, the same children paid a high physiological price for learning in one teaching system and learned without difficulty in another, which the authors read as a conflict between external management of learning and the learner’s own regulation [1].

Machine learning adopted the *order* and *dose* parts of pedagogy: curricula, self-paced learning, spaced repetition. It has adopted the protection of the learner, meaning state diagnosis and recovery on a threshold, only sparsely (Section 2). This note asks the smallest falsifiable version of that question.

2 Related work

Curriculum and pacing. Bengio et al. [4] order examples by difficulty. The BabyLM Challenge findings [5] are a negative result: curriculum-based submissions did not beat gains from architecture and training length. Pedagogical order does not win automatically.

Replay. Experience replay is a strong continual-learning baseline [6], even with tiny episodic memories [7]. We use the metrics of GEM [8] and the scenarios of van de Ven and Tolias [9].

When and what to replay. Maximally interfered retrieval chooses *what* to replay by predicted interference [10]. Task-free continual learning triggers consolidation from loss-surface signals [11]. Both are close in spirit to a state trigger.

Homeostasis and plasticity. Kim et al. [12] let a meta-model control regularisation strength. Jin [13] proposes homeostatic continual learning. Dohare et al. [14] show that continued SGD erodes plasticity, a “recovery debt” our setup does not measure.

3 Method

Data. We use scikit-learn’s bundled 8×8 digits (1,797 images), split per class 60/20/20 into train/val/test per seed. *Split-Digits* is class-incremental: tasks $\{0, 1\}, \dots, \{8, 9\}$ with a single 10-way head. *Permuted-Digits* is domain-incremental: 5 fixed pixel permutations of all 10 classes. We chose a dataset that needs no download. Replication on MNIST is the obvious next step.

Learner. An MLP 64–128–128–10 with ReLU, trained by plain SGD with learning rate 0.1 and batch 32 for $S = 300$ gradient steps per task (1,500 in total).

Fixed budget. Every arm makes exactly 1,500 gradient steps. A replay step (one batch from the memory of past tasks) *replaces* a current-task step. At the end of each task we store 20 training examples in the replay memory. At the *start* of each task, 10 more examples are set aside as a probe and are never trained on. Separate random streams choose current-task batches, replay batches, buffer exemplars and probes, so every arm stores identical exemplars.

Arms.

- **A:** no replay.
- **B:** replay every $N = 5$ steps (240 replay steps).
- **C (threshold):** every N steps the learner earns one replay credit and runs the diagnostic. The diagnostic is the mean rise of probe cross-entropy over all past tasks, relative to its value when each task ended. If the rise exceeds τ , a recovery phase starts. It replays while credits remain and the re-diagnosed rise stays above τ . By construction C never replays more than B.
- **Equal-count controls**, built from C’s realised replay steps in the same seed:
 - **B_match:** the same count, evenly spaced over the run.
 - **D_rand:** the same count, at random times.
 - **E_front:** C’s per-task counts at the start of each task.
 - **E_shuf:** C’s per-task counts at random inside each task.

- **E_even**: C’s per-task counts evenly spaced inside each task. This is an *oracle* control: it borrows the sensor’s per-task counts, so it isolates within-task timing and is not a deployable rule. It was added after earlier results were seen.
- **F_lin**: a sensor-free rule. The same total, split across tasks in proportion to the number of past tasks (1:2:3:4) and evenly spaced within each task.

Metrics. Final average test accuracy over the 5 tasks, backward transfer BWT [8], replay steps used, and diagnostic forward passes.

Protocol. In the main table, τ is tuned on the validation split with seeds 100–104 and reported on test with seeds 0–4. The cost curve uses a fixed τ set $\{0.5, 0.75, 1, 1.5, 2\}$. Three hypotheses were pre-registered. After two instrument fixes, the confirmatory run uses fresh seeds 20–29, which no version of the code had seen. That run’s code hash is recorded in `t2.sha1`. Seeds 0–9 and 10–19 are exploratory.

Red probe. With $\tau = +\infty$, C must reproduce A bit-for-bit. With $\tau = -\infty$, it must reproduce B bit-for-bit. Both hold for every seed and both benchmarks.

4 Results

Tuned threshold versus schedule. Validation tuning picks $\tau = 0.2$ (Split) and $\tau = 0.1$ (Permuted). At those values the trigger fires at almost every check and spends the whole budget (Table 1: 240 replay steps in every seed), so C degenerates into B. When the budget cap is generous and the objective is accuracy alone, a threshold has nothing to save.

Table 1: Tuned threshold vs schedule (τ tuned on val, seeds 0–4, test). All arms make 1,500 gradient steps. “diag” is the extra cost of the diagnostic: forward passes on the 10–40 held-out probe images.

bench	arm	final acc (mean $\pm$ sd)	BWT (mean $\pm$ sd)	replay steps	diag forwards
perm	A	0.503 $\pm$ 0.011	−0.552 $\pm$ 0.014	0	0
perm	B	0.776 $\pm$ 0.033	−0.206 $\pm$ 0.037	240	0
perm	C	0.774 $\pm$ 0.028	−0.209 $\pm$ 0.035	240	315
perm	B_match	0.779 $\pm$ 0.036	−0.201 $\pm$ 0.042	240	0
perm	D_rand	0.775 $\pm$ 0.028	−0.203 $\pm$ 0.029	240	0
split	A	0.197 $\pm$ 0.003	−0.999 $\pm$ 0.002	0	0
split	B	0.902 $\pm$ 0.025	−0.109 $\pm$ 0.031	240	0
split	C	0.898 $\pm$ 0.030	−0.113 $\pm$ 0.033	240	326
split	B_match	0.884 $\pm$ 0.031	−0.133 $\pm$ 0.036	240	0
split	D_rand	0.863 $\pm$ 0.043	−0.156 $\pm$ 0.054	240	0

Cost curve on the confirmatory holdout. Table 2 reports seeds 20–29. Every control uses exactly C’s replay count in the same seed. Accuracies are means over 10 seeds; each difference is a paired mean, with the number of seeds (of 10) in which C was higher in brackets. Under a two-sided sign test, 10/10 gives $p \approx 0.002$, 9/10 gives $p \approx 0.02$ and 8/10 gives $p \approx 0.11$. With ten comparison points per contrast and no correction, one 8/10 result per contrast is expected by chance; only 9/10 and 10/10 points carry weight on their own.

Table 2: Cost curve, confirmatory fresh holdout (seeds 20–29, test, fixed τ set). Mean accuracy; differences are paired means with C’s win count of 10 in brackets.

bench	τ	replay	C	B_match	E_shuf	E_even	F_lin	C–B_match	C–E_shuf	C–E_even	C–F_lin
split	0.5	187	0.859	0.825	0.837	0.852	0.855	+0.034 (10)	+0.022 (7)	+0.007 (6)	+0.004 (6)
split	0.75	137	0.827	0.789	0.817	0.831	0.825	+0.038 (8)	+0.009 (7)	−0.005 (5)	+0.002 (5)
split	1.0	85	0.816	0.738	0.778	0.808	0.800	+0.077 (10)	+0.038 (8)	+0.007 (7)	+0.016 (6)
split	1.5	51	0.685	0.621	0.632	0.732	0.709	+0.064 (9)	+0.054 (6)	−0.046 (3)	−0.024 (4)
split	2.0	36	0.568	0.530	0.538	0.635	0.629	+0.038 (8)	+0.030 (6)	−0.067 (0)	−0.062 (1)
perm	0.5	225	0.765	0.759	0.764	0.764	0.774	+0.006 (6)	+0.001 (6)	+0.001 (6)	−0.009 (2)
perm	0.75	145	0.759	0.737	0.758	0.754	0.755	+0.021 (10)	+0.000 (3)	+0.005 (6)	+0.003 (5)
perm	1.0	70	0.723	0.688	0.717	0.719	0.719	+0.035 (8)	+0.007 (6)	+0.005 (7)	+0.004 (6)
perm	1.5	14	0.656	0.615	0.642	0.647	0.633	+0.041 (9)	+0.014 (7)	+0.009 (8)	+0.023 (7)
perm	2.0	7	0.594	0.563	0.596	0.592	0.587	+0.031 (8)	−0.002 (4)	+0.002 (6)	+0.007 (7)

Exploratory seeds. The same curve on seeds 10–19 (seen once under the superseded v2 instrument) and on seeds 0–9 gives the same picture; both tables are in the repository as the two exploratory `curve_seeds` files.

Two further controls. Table 3 shows `E_front` and `D_rand` on the holdout. `E_front`, which places each task’s replay at the start of the task, sits at the no-replay level on Split (0.197 at every τ , against 0.197 for A): replay placed before the interference is wasted. `D_rand`, random global timing, is no better than uniform.

Table 3: `E_front` and `D_rand` against C on the confirmatory holdout (seeds 20–29, mean accuracy).

bench	τ	E_front	D_rand	C
split	0.5	0.197	0.825	0.859
split	1.0	0.197	0.699	0.816
split	2.0	0.197	0.486	0.568
perm	0.5	0.570	0.761	0.765
perm	1.0	0.541	0.691	0.723
perm	2.0	0.522	0.558	0.594

Pre-registered hypotheses (Table 2).

- **H1** (C beats B_match on Split at τ 0.5–1.0 in $\geq 8/10$ seeds): **confirmed** (10, 8 and 10 of 10; +3.4, +3.8 and +7.7 points). H1 compares C with *globally* uniform replay, so it supports the claim that C’s allocation across stages is better. It does not show that state-timing matters on its own.
- **H2** (within-task timing does not matter: $|C - E_shuf| < 0.02$ with 3–7 wins): **rejected on Split** at 4 of 5 τ values (+2.2 to +5.4 points). It **holds on Permuted** at all 5.
- **H3** (C does not beat the sensor-free rule F_lin: $\leq 7/10$ wins at every point): **confirmed at all 10 points** (C higher in 1–7 of 10). At $\tau = 2.0$ on Split, F_lin is better by 6.2 points (C higher in 1 of 10).

Exploratory reading of the H2 rejection. `E_even` (the oracle) uses C’s per-task counts, spaced evenly instead of at random. It matches C at medium budgets and beats it at the two smallest (−4.6 and −6.7 points; C higher in 3 and 0 of 10 seeds). The gap between C and `E_shuf` is therefore consistent with random placement clumping replay steps together, and not with the sensor choosing good moments.

5 Negative results and what they mean

1. **A threshold tuned for accuracy alone turns into a schedule.** With a cap of 240 replay steps, the best validation τ spends the whole cap (Table 1). The saving that motivates threshold triggering can only appear when cost enters the objective. We did not optimise a cost-accuracy trade-off and only report the curve.
2. **The diagnostic only does dosage.** Every advantage of the sensor over a uniform schedule is reproduced by a fixed, sensor-free rule that gives later stages more recovery (F_lin). The oracle control (E_even) shows that the sensor’s moment-to-moment timing adds nothing beyond its per-stage counts and costs accuracy at small budgets. In pedagogical terms, the transferable part of the health-preserving lesson is its dosing rule, which a teacher can apply without measuring the class.
3. **Recovery placed before the interference is wasted.** The intuitive schedule of consolidating immediately after moving to a new topic is the worst arm (Table 3). Recovery has to be spread across the period in which the new material interferes with the old.
4. **Two earlier versions were wrong, and both were caught by external review.** In v1 the probe was drawn from already-trained examples. In v2 a shared random stream gave arms with different replay timing different buffer exemplars. The v1 fix moved tuned- τ mean accuracy of C on Split by +2.1 points (seeds 0–4); the v2 fix moved mean C accuracy on Split at $\tau = 2.0$ by +9.5 points (seeds 10–19). The conclusion in item 2 survived both. Because seeds 10–19 had been seen under v2, the confirmatory claim rests on fresh seeds 20–29 (addendum 2 of the pre-registration), where it replicates.
5. In line with the BabyLM negative result, a pedagogically motivated mechanism did not dominate a simple engineering rule.

6 Limitations

- The data and model are tiny: 8×8 digits, an MLP with two hidden layers, 5 tasks. Effects at this scale may not transfer; MNIST, CIFAR and larger models are untested.
- The probe is 10 extra stored examples per task, which the schedule arms do not use. The diagnostic also costs forward passes (reported) that the schedule arms do not pay.
- F_lin and E_even were designed after seeing earlier results. F_lin is tested by the pre-registered H3 on the fresh holdout. E_even is an exploratory oracle.
- Task boundaries are known to every arm. A sensor’s real advantage could appear only when boundaries are unknown, which we did not test.
- In Permuted-Digits the same images appear under different permutations in different tasks, so “held out” is per task, not global.
- The learner-state vector that the pedagogical tradition works with (gradient instability, representation damage, calibration, plasticity loss, recovery debt) is reduced here to a single loss-rise signal.
- Each comparison uses 10 seeds and a sign test with no correction for multiple comparisons across 10 points.

7 Conclusion

On a deliberately small benchmark, recovery triggered by a diagnosed loss rise beats recovery spread evenly over the whole run at the same cost on Split-Digits, which confirms the pre-registered H1 on a fresh holdout; on Permuted-Digits the same contrast is reliable only at $\tau \geq 0.75$. That gain is a matter of *dosage*: how much recovery each learning stage gets, spread evenly within the stage. A sensor-free rule (“more review as more material accumulates”) reproduces it at every budget we tried.

For the health-preserving programme the first number is therefore a split verdict. The *dosage* principle transfers and can be measured. The *diagnostic trigger* has not yet earned its keep over a simple rule. The next tests are (a) MNIST and CIFAR at realistic scale, (b) an objective that prices replay, so that a threshold can save, (c) richer state signals than loss rise alone, and (d) streams with unknown task boundaries, where a fixed dosage rule might fail to follow shifts that a sensor might follow.

Reproducibility

`python t2.py --all` (about 8 minutes on one CPU core), `python t2.py --curve --seeds 20-29` (confirmatory), `--seeds 10-19` and `--seeds 0-9` (exploratory; a few minutes each). Code, raw CSVs, the pre-registration with its addenda, the Codex review rounds that found the v1 and v2 defects, and both superseded versions are at <https://github.com/tonydz/machine-pedagogy> (folder `t2-threshold-replay`).

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